

HARSH JANNAWAR

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EDUCATION

University of Washington

Master of Science in Cybersecurity Engineering, GPA: 3.93/4

09/2024 – 06/2026 | Seattle, WA

Symbiosis Skills and Professional University

Bachelor of Technology: CSIT (Cybersecurity), GPA 3.65/4

08/2020 – 05/2024 | Pune, India

PROFESSIONAL EXPERIENCE

AI Security Engineer

Cyber Alliance

09/2026 – Present | Raleigh, North Carolina

- Built an org-specific threat-intelligence pipeline ingesting 12 scheduled feeds (CISA KEV, EPSS), geo-tagging indicators, and filtering threats by each organization's region, industry, and tech stack with plain-English impact summaries.
- Built an agentic security assistant with a 7-tool calling loop grounded in live scan findings, posture, and assessment data, hardening it against hallucination with severity-sorted, size-capped tool output and against prompt injection.
- Delivered full-stack vulnerability scanning with web, network, and ZAP active scans, Semgrep and Trivy source-code scanning, and a NIST-based weighted risk assessment, generating Executive, Technical, and Compliance PDF reports from a shared pipeline.
- Built org-scoped RBAC (Owner/Admin/Member across 13 capabilities) enforced server-side at 26 endpoints with Auth0, role-based MFA, and fail-secure step-up MFA before irreversible actions, closing privilege-escalation gaps and covered by ~97 tests.
- Took the platform from nothing to production on AWS with ~210 Terraform/Pulumi-managed resources (VPC, ECS Fargate, RDS, KMS) and GitHub Actions CI/CD using OIDC with no stored keys, shipping 80+ verified releases and authoring 195 of 236 merged PRs across 8 repos.
- Built GitHub Actions build, push-to-ECR, and deploy pipelines that authenticate to AWS through OIDC with no stored keys, auto-triggering ECS rollouts on promotion. Added branch protection, Codeowner, and a main-to-prod release model across repos, and stood up CI/CD for services that had none, shipping 80+ verified production releases with rollback safety.

AI Cybersecurity Engineering Intern

SecureAIs

06/2025 – 08/2025 | California

- Developed a PII detection and redaction pipeline using Presidio, regex-based validation, and packet analysis to protect sensitive data across 10k test samples, improving sensitive data protection accuracy from 65% to 95%.
- Built and tested end-to-end authentication flows across engineering services, including secure login, session handling, token validation, role-based access control, and protected API routes to reduce unauthorized access risk across application workflows by 30%.
- Conducted adversarial testing and secure code reviews across AI platforms to uncover 20+ critical vulnerabilities including authentication bypass, prompt injection, and token leakage prior to production release, delivering CWE-mapped remediation guidance to engineering within 48 hours.
- Implemented static analysis and dynamic analysis by integrating SAST, DAST, dependency scanning, secret detection, and Control Gating into CI/CD pipelines, reducing security review cycles by 60% and preventing 12 high-risk releases from deploying until fixes were verified.

Security Analyst Intern

SecureThings

06/2023 – 05/2024 | India

- Built an event-driven scanning platform using Dockerized asynchronous Python, Nmap, Nuclei, CVE, EDBID's and open-source services deployed via Amazon ECR and ECS, saved in S3 and MongoDB, cutting asset analysis time from 9+ hours to under 30 minutes.
- Performed application and infrastructure security risk assessments using Nessus, Nmap, and Burp Suite to triage 15+ high-severity vulnerabilities with validated exploits, reducing time-to-remediate by 35% through developer-ready reproduction steps.
- Designed backend risk-scoring logic to prioritize findings based on exploitability, exposed services, affected technologies, severity, and business context, improving remediation prioritization accuracy by 40% and cutting false positives by 25%.
- Strengthened AWS and Azure environments by improving IAM permissions, container security controls, secrets handling, and exposed service configurations, remediating 25+ cloud and application security gaps and reducing public exposure findings by 50%.

CORE COMPETENCIES & TECHNICAL EXPERTISE

Technical Skills: Secure SDLC, Threat Modeling, Secure Code Review, Penetration Testing, Vulnerability Management, Authentication & Authorization, Cloud Security, Detection Engineering, Prompt Injection Defense, LLM Red Teaming, Agentic AI Security, MCP Security, Policy-as-Code

Programming: Python, TypeScript, Next.js, SQL, JavaScript, Node.js, Bash Scripting

Frameworks and Database: NIST AI RMF, NIST CSF, MITRE ATLAS, MITRE ATT&CK, OWASP LLM Top 10, ISO/IEC 27001, ISO/IEC 42001, SOC 2.

Tools: Burp Suite, OWASP ZAP, Nuclei, Nmap, Semgrep, CodeQL, Trivy, Gitleaks, Splunk, AWS (IAM, ECS, S3, KMS, CloudTrail, WAF), Pulumi, Terraform, GitHub Actions, Docker.

Certifications: CompTIA Security+, CompTIA Pentest+ (THM), DevSecOps, TCM Practical Ethical Hacking, Oracle Cloud Infrastructure Foundations

PROJECTS

Inline Proxy - Security Gateway for AI Agents (Python, asyncio)

08/2026

- Built a transparent MCP proxy that enforces role-aware, per-tool policy (allow, block, redact, rewrite, quarantine, human-approval) on every agent tool call via a pluggable asyncio pipeline and layered YAML policy packs, running fail-closed with 600+ CI tests and sub-millisecond p95 latency.
- Added prompt-injection and data-exfiltration defenses for agentic AI workflows, including PII/secret redaction, taint tracking, per-session risk scoring, and an LLM behavioral anomaly monitor, blocking a full prompt-injection exfiltration chain end to end in automated attack simulations against a live MCP server.
- Implemented policy-as-code CI gating for tool access and data egress, running golden-test regression suites and blast-radius diffs that block a merge when a policy change silently weakens enforcement, backed by deterministic, git-reviewable YAML rules

- Aegis LLM - Automated LLM Security Testing Suite** 

03/2026
- Engineered an adversarial red-teaming framework mapped to the OWASP LLM Top 10 that probes LLM applications for prompt injection, jailbreaks, and rate-limit DoS across 6 attack classes, using an LLM-as-a-judge engine to score injection success and generate reproducible reports, reducing manual test effort by 70%.
 - Built advanced attack modules, including a genetic algorithm that evolves failed jailbreak prompts and flags guardrail bypasses, a differential scan against a baseline model, and a recon scan that fingerprints the backend model, detects RAG, and enumerates connected tools.

- Cloud Security Detection, AI Explainer & Compliance Reporting Engine** 

06/2026
- Built a cloud detection engine with 10 configurable modules over CloudTrail, WAF, and network flow logs, with composite entity risk scoring (0 to 100+) that aggregates weighted alerts so repeated suspicious behavior escalates into Critical risk, increasing true-positive triage yield by 45%.
 - Automated AI-assisted reporting that generates technical runbooks, executive summaries, and customer-facing explanations, mapping each finding to MITRE ATT&CK, OWASP Top 10, PCI DSS 4.0, and GLBA via verified static mappings for compliance, cutting report turnaround time by 60% and supporting UAT sign-off in 3 cycles.

ACHIEVEMENTS & TECHNICAL CONTRIBUTIONS

- Ranked Top 1% on TryHackMe**

Hands-on experience across DevSecOps, penetration testing, red team, and blue team paths and competed in CTF challenges.
- Published cybersecurity and technical blogs on Medium** 

- Published Research: AI Security Compliance and Testing Framework for LLM Systems** 

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